



Block Diagram of a Computer System

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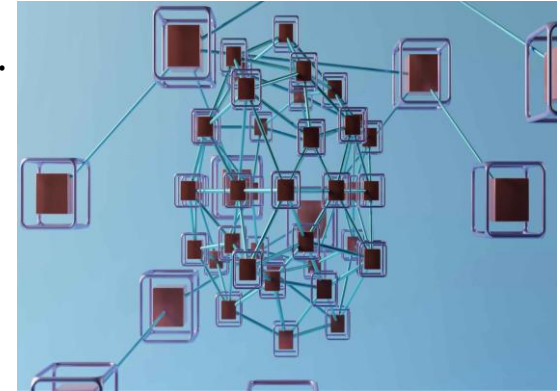
- Block Diagram of a Computer System
- 4 core Functional units :-
 - Input unit
 - Output unit
 - Memory unit
 - CPU
- Bus System : Description of data bus, address bus, control bus
- Component interaction
- Summary

Block Diagram of a Computer System



A block diagram provides a high-level, simplified visual representation of a computer's main components and their interconnections. It's an essential tool for understanding the fundamental architecture and operational flow of any computing device.

- Illustrates how key units like input, CPU, memory, and output interact.
- Helps visualize the path data takes as it's processed and transformed.
- Highlights the role of various buses in facilitating communication.



By breaking down the complex internal workings into manageable blocks, it demystifies how computers efficiently execute tasks, from simple data entry to complex computations.

The Four Core Functional Units

At the heart of every computer system are four functional units that collaborate to perform all operations.

1

Input Unit

Devices that feed raw data and instructions into the system for processing.

2

Central Processing Unit (CPU)

The "brain" responsible for executing instructions and processing data.

3

Memory Unit

Stores data and instructions, providing quick access for the CPU.

4

Output Unit

Presents processed results to the user in a human-understandable format.

These units form the fundamental framework that allows a computer to process simple & complex tasks.

An input unit of a computer system performs the following functions:

1. It accepts (or reads) instructions and data from outside world
2. It converts these instructions and data in computer acceptable form
3. It supplies the converted instructions and data to the computer system for further processing

"Without a robust input unit, the most powerful CPU is useless, as it would have no data to process."

Common examples include keyboards, mice, microphones, scanners, and touchscreens. These devices are crucial for any user interaction.



An output unit of a computer system performs the following functions:

1. It accepts the results produced by the computer, which are in coded form and hence, cannot be easily understood by us
2. It converts these coded results to human acceptable (readable) form
3. It supplies the converted results to outside world



Output unit is vital for user interaction, as it provides the feedback and results of all computational tasks.

The memory unit of a computer system holds (or stores) the following:

1. Data and instructions required for processing (received from input devices)
2. Intermediate results of processing
3. Final results of processing, before they are released to an output device

Primary Memory (RAM)

Temporary & Fast: Volatile storage (data is lost when power is off) used for actively running programs and data the CPU is currently using. It offers very fast access times.

Secondary Memory (SSD, HDD)

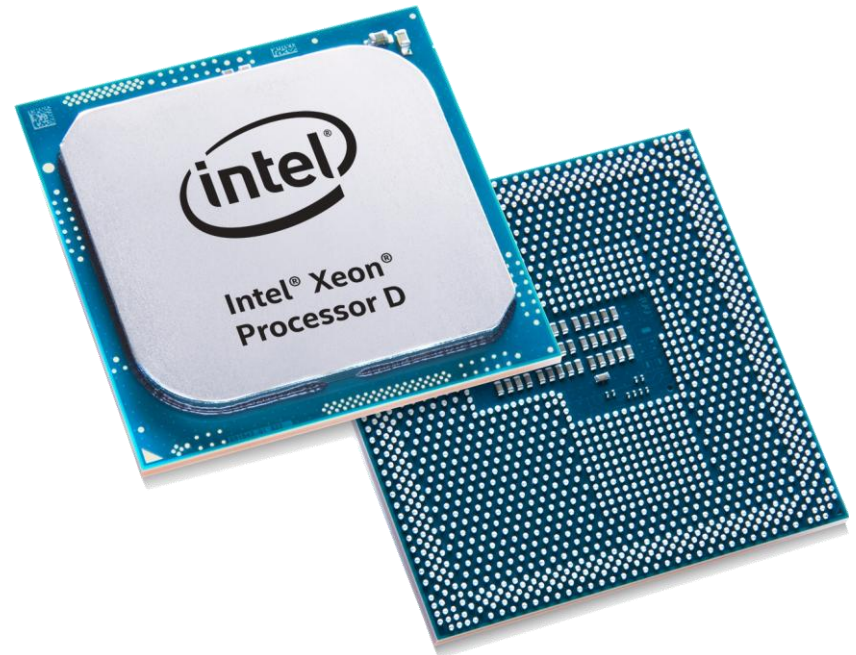
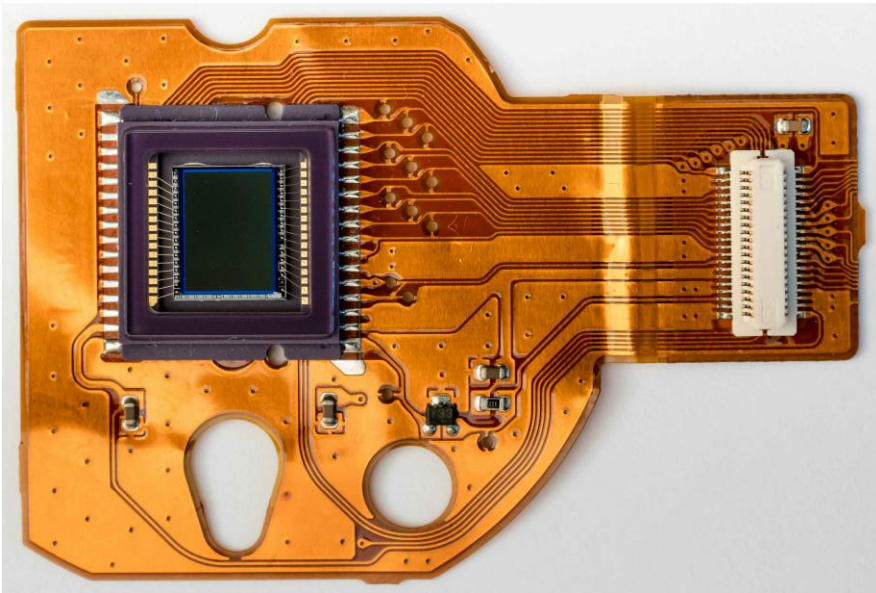
Permanent & Slower: Non-volatile storage used for long-term storage of files, operating systems, and applications. Data persists even when the computer is turned off.

The seamless interaction between primary and secondary memory ensures that the CPU always has the necessary data at hand, optimizing overall system performance.



Often referred to as the "brain" of the computer, the CPU is responsible for executing instructions, performing calculations, and managing the overall operation of the system. It consists of two main sub-units:

- **Arithmetic Logic Unit (ALU):** Handles all arithmetic operations (addition, subtraction, etc.) and logical comparisons (e.g., greater than, equal to).
- **Control Unit (CU):** Directs and coordinates the flow of data within the CPU and between the CPU, memory, and input/output (I/O) devices. It fetches instructions, decodes them, and then directs other components to perform specific tasks.



Buses are **electrical pathways** that allow data, addresses, and control signals to travel between different components of the computer system. They are the "highways" facilitating communication.

Data Bus

Transfers the actual data being processed or stored between the CPU, memory, and I/O devices. It's bi-directional.

Address Bus

Carries memory addresses to specify the location where data should be read from or written to. It's uni-directional (from CPU memory/I/O).

Control Bus

Transmits control signals from the Control Unit to other components, coordinating all operations (e.g., read/write signals, interrupt requests).

These buses work in unison to ensure precise and synchronized data transfer across the entire system.

How Components Interact?

To illustrate the seamless interaction of these units, consider a simple example: typing a character on a keyboard and seeing it appear on the screen.

01

Input & Conversion

User presses a key on the keyboard. The Input Unit converts this physical action into a machine-readable binary code.

03

CPU Processing

The **CPU's Control Unit** fetches the data from memory using the **Address** and **Data Buses**. The **ALU** may perform any necessary processing (e.g., validating the character).

02

Storage in Memory

This binary data is sent via the **Data Bus** to the **Memory Unit** (RAM), where it is temporarily stored. The **Address Bus** tells memory exactly where to place it.

04

Output Display

The processed data is sent back through the **Data Bus** to the **Output Unit** (monitor). The Output Unit converts the binary data into a visual signal, and the character appears on the screen.

- The block diagram reveals the simplicity behind complex computing. It underscores how the Input, CPU, Memory, and Output units, interconnected by the essential Data, Address, and Control Buses, work in perfect harmony to execute every task.
- The **Control Unit** acts as the conductor, ensuring precise timing and sequence for all operations.
- Understanding this foundation is key to appreciating the power and efficiency of modern computing.

Q1. A block diagram of a computer system primarily shows:

- a) Internal circuitry details
- b) High-level components and their interaction
- c) Operating system design
- d) Software applications

Answer – b) High-level components and their interaction



Q2. Which of the following is **NOT** a core functional unit of a computer system?

- a) Input Unit
- b) Output Unit
- c) Arithmetic Unit
- d) Memory Unit

Answer – c) Arithmetic Unit



Q3. The Input Unit is essential because:

- a) It stores all data permanently
- b) It feeds raw data and converts it into machine-readable form
- c) It controls execution of instructions
- d) It displays results to the user?

Answer – b) It feeds raw data and converts it into machine-readable form

Q4. Which of the following is an example of an input device?

- a) Monitor
- b) Keyboard
- c) Printer
- d) Speaker

Answer – b) Keyboard

Q5. The Output Unit's main function is:

- a) Storing temporary data
- b) Converting results into human-readable form
- c) Performing calculations
- d) Fetching instructions

Answer – c) Converting results into human-readable form



Q6. Which type of memory is **volatile** and loses its contents when power is off?

- a) RAM
- b) ROM
- c) SSD
- d) HDD

Answer – a) RAM

Q7. The CPU consists of:

- a) ALU and Memory Unit
- b) ALU and Control Unit
- c) Input Unit and Output Unit
- d) Control Bus and Address Bus

Answer – **b) ALU and Control Unit**

Q8. Which bus is **bi-directional** and transfers actual data between CPU, memory, and I/O devices?

- a) Address Bus
- b) Control Bus
- c) Data Bus
- d) System Bus

Answer – **c) Data Bus**

Q9. The Control Unit in a CPU is responsible for:

- a) Performing arithmetic calculations
- b) Directing and coordinating data flow
- c) Storing data permanently
- d) Displaying output

Answer – **b) Directing and coordinating data flow**



Q10. In the example of typing a character on the keyboard, the data first goes to:

- a) CPU
- b) Output Unit
- c) Memory Unit
- d) Control Bus

Answer – **c) Memory Unit**